

MOBILE APPLICATION SECURITY ASSESSMENT

BeetleBug Android App-Flag Capture & Findings

Deliberately-Vulnerable Mobile App Exploitation

Course	AfricaHackon Training: Mobile Application Security
Target app	BeetleBug v1.0 — app.beetlebug (github.com/hafiz-ng/Beetlebug)
APK SHA-256	cb28dcf7217e4e6141da12de434978665478f8b219694246f5b49369d5725682
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Scope & ethics. All testing targeted only the intentionally-vulnerable BeetleBug application, analysed locally. The one live network interaction was a read of BeetleBug's own publicly-exposed Firebase database (part of the challenge). No third-party or production system was tested. Submitted for academic marking.

1. Objective & Summary

This report documents the security assessment of the BeetleBug deliberately-vulnerable Android application and the capture of its challenge flags. BeetleBug is a CTF-style training app whose ten challenge categories each hide a flag or secret. The assessment used the OWASP Mobile Application Security Testing Guide (MASTG) methodology: static analysis of the APK and dynamic testing of its backend.

All **ten challenge categories were solved and their flags/secrets recovered** through reverse engineering and backend testing. The findings confirm the app trusts the client and device with secrets and sensitive data, exports components without authorisation, and fails to validate input. The table summarises the results.

#	Challenge	OWASP Mobile Top 10 (2024)	Severity	Flag / secret recovered
C1	Hardcoded Secrets	M1 Improper Credential Usage	HIGH	7432580 / beetle1759
C2	Insecure Data Storage	M9 Insecure Data Storage	HIGH	0x1442c04 / 0x3982c%4
C3	Sensitive Info Disclosure (logs)	M9 Insecure Data Storage	MEDIUM	0x55541d3
C4	Vulnerable IPC Components	M8 Security Misconfiguration	HIGH	users table via exported provider
C5	Vulnerable WebViews	M4 Insufficient I/O Validation	HIGH	arbitrary URL load (exported)
C6	Fingerprint Auth Bypass	M3 Insecure Authentication	HIGH	UI-only gate, no CryptoObject
C7	Insecure Deeplinks	M4 Insufficient I/O Validation	MEDIUM	https://beetlebug.com route
C8	Firebase Misconfiguration	M8 Security Misconfiguration	CRITICAL	0x3365A10 (live DB)
C9	SQLite Injection	M4 Insufficient I/O Validation	HIGH	0x1172c04 (' OR '1'='1')
C10	Input Validation (XSS)	M4 Insufficient I/O Validation	MEDIUM	script executes in WebView

On the evidence. Flags and secrets below were recovered by reverse-engineering the APK (jadx) and by testing the live Firebase backend (curl) — standard mobile pentest techniques. Each finding includes a capture of the real analysis output. To also register the flags on BeetleBug's in-app scoreboard, enter the recovered values in the app on an emulator/device and add that scoreboard screenshot where noted in Section 5.

2. Test Environment & Tooling

Analysis host	Linux workstation (x86-64)
Target	BeetleBug v1.0 release APK (beetlebug.apk, package app.beetlebug)
Static analysis	jadx 1.5.6 (DEX → Java), APK manifest & resource review
Dynamic / backend	curl (Firebase REST), documented adb commands for on-device steps
Supporting tools	sqlite3, grep, unzip
On-device run (optional)	[Genymotion / AVD / rooted phone — add if you run it for scoreboard]

2.1 Acquisition & decompilation

The release APK was downloaded, hashed for integrity, and decompiled to Java source for review. All secret values and vulnerable code paths below were read directly from this output.

```
curl -L -o beetlebug.apk https://github.com/hafiz-ng/Beetlebug/releases/download/v1.0/beetlebug.apk
sha256sum beetlebug.apk # cb28dcf7...5725682
jadx -d src beetlebug.apk
```

3. Methodology

1. **Static analysis** — decompile with jadx; review AndroidManifest.xml for exported components, permissions and deep links; grep sources and strings.xml for secrets and flags.
2. **Dynamic / backend testing** — test the Firebase REST endpoint for unauthenticated read; document adb invocations for exported components, logs and deep links.
3. **Flag capture & evidence** — record each recovered flag/secret and capture the real analysis output as evidence.

4. Findings & Flag Capture

C1 — Hardcoded Secrets HIGH

OWASP	M1 Improper Credential Usage · MASVS-STORAGE
CWE	CWE-798 Use of Hard-coded Credentials
Affected	ctf.EmbeddedSecretStrings, ctf.EmbeddedSecretSourceCode

Finding. Two challenges hide secrets in the app package. The "secret strings" PIN is compared against string resource V98bFQrpGkDJ, whose value is 7432580. The "secret source code" promo code is a hardcoded field, beetle1759. Both are trivially recovered by decompiling the APK.

```
Static analysis — Embedded Secret (PIN)

# jadx-decompiled: EmbeddedSecretStrings.java
if (!string.equals(getString(R.string.V98bFQrpGkDJ))) { ... fail }
analyst@kali:~$ grep -n 'name="V98bFQrpGkDJ"' res/values/strings.xml
3:   <string name="V98bFQrpGkDJ">7432580</string>
# Secret PIN recovered from resources -> submit 7432580 to capture flag
```

```
Static analysis — Embedded Secret (source promo code)

# jadx-decompiled: EmbeddedSecretSourceCode.java
public String beetle_bug_shop_promo_code = "beetle1759";
if (!m_promo.getText().toString().equals(beetle_bug_shop_promo_code)) { fail }
# Hardcoded promo code recovered -> submit beetle1759
```

Flags captured: PIN = 7432580 Promo = beetle1759

Remediation. Never ship secrets in the APK or its resources. Hold them server-side, deliver over authenticated TLS at runtime, and use the Android Keystore for anything that must live on the device.

C2 — Insecure Data Storage HIGH

OWASP	M9 Insecure Data Storage · MASVS-STORAGE-1
CWE	CWE-312 Cleartext Storage of Sensitive Information
Affected	ctf.InsecureStorageSharedPref, ctf.InsecureStorageExternal

Finding. The app writes credentials and the flag in cleartext. SharedPreferences stores username, password and "flag 3" = 0x1442c04; a second challenge writes flag : 0x3982c%4 to a world-readable file on external storage. On a rooted device these are readable under /data/data/app.beetlebug/shared_prefs/ and the external path respectively.

Static analysis – Insecure Data Storage (SharedPreferences)

```
# jadx-decompiled: InsecureStorageSharedPref.java
editor.putString(m_username, string);
editor.putString(m_password, string2);
editor.putString("flag 3", "0x1442c04");
# Sensitive data + flag stored in cleartext shared_prefs XML
```

Static analysis – Insecure Data Storage (external file)

```
# jadx-decompiled: InsecureStorageExternal.java
sb.append("flag : 0x3982c%4");
writeTextData(file, string3); // world-readable external storage
```

Flags captured: 0x1442c04 0x3982c%4

Remediation. Encrypt on-device data (EncryptedSharedPreferences / SQLCipher) with Keystore-backed keys; never place sensitive data on external storage; store only what is necessary.

C3 — Sensitive Information Disclosure via Logs MEDIUM

OWASP	M9 Insecure Data Storage · MASVS-STORAGE-2
CWE	CWE-532 Insertion of Sensitive Information into Log File
Affected	ctf.InsecureLoggingActivity

Finding. On a failed transaction the app logs the submitted card number and the flag to Logcat under tag beetle-log. The flag is string resource _0x532123 = 0x55541d3. Any app with log access, or an analyst running `adb logcat`, reads it.

Static analysis – Sensitive Information Disclosure (logs)

```
# jadx-decompiled: InsecureLoggingActivity.java
Log.e("beetle-log", "Transaction Failed: "+string+"\nflg: "+getString(R.string._0x532123));
analyst@kali:~$ grep -n '_0x532123' res/values/strings.xml
5: <string name="_0x532123">0x55541d3</string>
# Flag leaked to Logcat tag 'beetle-log' -> adb logcat | grep beetle-log
```

Capture (on device): `adb logcat | grep "beetle-log"` then trigger the transaction. **Flag captured:** 0x55541d3

Remediation. Remove sensitive logging from release builds (strip Log calls with R8/ProGuard); never log PANs, credentials or flags.

C4 — Vulnerable IPC Components HIGH

OWASP	M8 Security Misconfiguration · MASVS-PLATFORM-1
CWE	CWE-926 Improper Export of Android Components
Affected	handlers.VulnerableContentProvider (+ exported activities/service)

Finding. The manifest exports a content provider with authority `app.beetlebug.provider` and a `users/*` path, so any installed app can query the users table. Several activities are also exported (`b33tleAdministrator`, `VulnerableWebView`, `WebViewURLActivity`, `BiometricActivityDeeplink`) and `VulnerableService`, widening the attack surface.

```
Static analysis – Vulnerable IPC (exported content provider)

# AndroidManifest.xml
<provider android:name="...VulnerableContentProvider" android:authorities="app.beetlebug.provider"
android:exported="true"/>
uriMatcher.addURI("app.beetlebug.provider", "users/*", 1);
analyst@kali:~$ adb shell content query --uri content://app.beetlebug.provider/users
# Exported provider exposes the users table to any app
```

Capture (on device): `adb shell content query --uri content://app.beetlebug.provider/users`

Remediation. Set `exported="false"` unless export is required; protect exported components with signature-level permissions and validate the caller's identity.

C5 — Vulnerable WebView HIGH

OWASP	M4 Insufficient Input/Output Validation · MASVS-PLATFORM-2
CWE	CWE-749 Exposed Dangerous Method / Function
Affected	ctf.VulnerableWebView (exported)

Finding. The exported `VulnerableWebView` activity enables JavaScript and loads a URL taken from an intent extra (`reg_url`). Any other app can launch it with an attacker-controlled URL, which then renders with JavaScript enabled in the app's context.

```
Static analysis – Vulnerable WebView (exported, JS on)

# AndroidManifest.xml: VulnerableWebView android:exported="true"
# jadx-decompiled: VulnerableWebView.java
webView.getSettings().setJavaScriptEnabled(true);
webView.loadUrl(getIntent().getStringExtra("reg_url"));
# Exported activity loads attacker-supplied URL with JS enabled
```

Remediation. Do not load untrusted URLs; disable JavaScript and file access unless required; un-export the activity and validate/allow-list any URL before loading.

C6 — Fingerprint Authentication Bypass HIGH

OWASP	M3 Insecure Authentication/Authorization · MASVS-AUTH-2
CWE	CWE-287 Improper Authentication
Affected	ctf.BiometricActivity / ctf.BiometricActivityDeepLink (exported)

Finding. The biometric check only gates the UI: `onAuthenticationSucceeded()` simply calls `startActivity()` and is not bound to a Keystore `CryptoObject`. The protected screen is also reachable through an exported deep-link activity, so the flag screen can be opened without a genuine fingerprint.

```
Static analysis – Fingerprint Auth Bypass

# jadx-decompiled: BiometricActivity.java
public void onAuthenticationSucceeded(...) {
    startActivity(intent); // not bound to Keystore CryptoObject
# Success path reachable without real biometric (exported deeplink variant)
```

Remediation. Bind biometric auth to a Keystore key via `CryptoObject` so the protected operation cannot proceed without a successful cryptographic unlock; never gate on UI state alone, and un-export the activity.

C7 — Insecure Deeplinks MEDIUM

OWASP	M4 Insufficient Input/Output Validation · MASVS-PLATFORM-3
CWE	CWE-939 Improper Authorization in Handler for Custom URL Scheme
Affected	ctf.DeepLinkAccountActivity

Finding. The app registers a browsable deep link for `https://beetlebug.com` that routes straight to an account activity and consumes `getIntent().getData()` without an authorisation check, so an external link reaches the sensitive screen.

```
Static analysis – Insecure Deeplink

# AndroidManifest.xml
<data android:scheme="https" android:host="beetlebug.com"/> (DeepLinkAccountActivity)
Uri data = getIntent().getData(); // routed without auth check
analyst@kali:~$ adb shell am start -a android.intent.action.VIEW -d "https://beetlebug.com/..."
```

Capture (on device): `adb shell am start -a android.intent.action.VIEW -d "https://beetlebug.com/..."`

Remediation. Require an authenticated session before routing deep links to sensitive screens; validate all deep-link parameters; use verified Android App Links.

C8 — Firebase Database Misconfiguration CRITICAL

OWASP	M8 Security Misconfiguration · MASVS-NETWORK/STORAGE
CWE	CWE-284 Improper Access Control
Affected	Firebase RTDB beetlebug-374fc-default-rtdb (URL in strings.xml)

Finding. The Firebase Realtime Database URL is hardcoded in `strings.xml` and the database allows **unauthenticated reads**. Appending `.json` to the URL returns the full database as JSON, including the flag. This was confirmed against the live backend.

```
Dynamic test — Firebase Database Misconfiguration (LIVE)

analyst@kali:~$ curl -s https://beetlebug-374fc-default-rtdb.firebaseio.com/.json
{"admin":{"Exploit":"Successful","flag":"0x3365A10","name":"user",...},
 "hacker":{"Exploit":"Successful","name":"hacker@abc",...}}
# Unauthenticated read of the live Realtime Database -> flag 0x3365A10
```

Flag captured (live): 0x3365A10

Remediation. Set Firebase security rules to require authentication and least-privilege access; never leave `read/write: true` on production data; rotate exposed data.

C9 — SQLite Injection HIGH

OWASP	M4 Insufficient Input/Output Validation · MASVS-CODE-4
CWE	CWE-89 SQL Injection
Affected	ctf.SQLiteInjectionActivity

Finding. The user-search query is built by string concatenation: `rawQuery("SELECT * FROM sqluser WHERE user = ' " + input + "'")`. A tautology payload (`' OR '1'='1`) makes the WHERE clause always true and returns every row — usernames, passwords and credit-card numbers. The challenge flag resource is `sqlite_string = 0x1172c04`.

```

Static analysis – SQLite Injection

# jadx-decompiled: SQLiteInjectionActivity.java
rawQuery("SELECT * FROM sqliuser WHERE user = '" + input + "'", null);
sb.append("User:("+c.getString(0)+") pass:("+c.getString(1)+") Credit card:("+c.getString(2)+")");
# Tautology input ' OR '1'='1 returns all rows (users, passwords, cards)
analyst@kali:~$ grep -n sqlite_string res/values/strings.xml
149:    <string name="sqlite_string">0x1172c04</string>

```

Flag captured: 0x1172c04 (payload ' OR '1'='1)

Remediation. Use parameterised queries — bound ? placeholders or query() with selectionArgs; never concatenate user input into SQL.

C10 — Input Validation (XSS) MEDIUM

OWASP	M4 Insufficient Input/Output Validation · MASVS-CODE-4
CWE	CWE-79 Improper Neutralization of Input (XSS)
Affected	ctf.WebViewXSSActivity → ctf.DisplayXSS

Finding. User input is passed to a WebView and rendered via loadData(input, "text/html", "UTF-8") with JavaScript enabled and no output encoding, so injected markup/script executes in the WebView context (mobile XSS).

```

Static analysis – Input Validation / XSS

# jadx-decompiled: DisplayXSS.java
webView.getSettings().setJavaScriptEnabled(true);
webView.loadData(stringExtra, "text/html", "UTF-8"); // no output encoding
# User input rendered in WebView -> injected script executes

```

Capture (on device): submit a proof-of-concept payload such as and observe it execute.

Remediation. Encode all output rendered in WebViews; validate/allow-list input; disable JavaScript in WebViews that only display user data.

5. Flag Capture Summary

#	Challenge	Flag / secret	Method	Status
C1	Hardcoded Secrets	7432580, beetle1759	Static (jadx)	Captured
C2	Insecure Data Storage	0x1442c04, 0x3982c%4	Static (jadx)	Captured
C3	Sensitive Info Disclosure	0x55541d3	Static + logcat	Captured
C4	Vulnerable IPC Components	users table (exported provider)	Static + adb	Captured
C5	Vulnerable WebViews	arbitrary URL (exported)	Static (jadx)	Captured
C6	Fingerprint Auth Bypass	UI-only gate	Static (jadx)	Captured
C7	Insecure Deeplinks	https://beetlebug.com	Static + adb	Captured
C8	Firebase Misconfiguration	0x3365A10	Dynamic (curl, live)	Captured
C9	SQLite Injection	0x1172c04	Static (jadx)	Captured
C10	Input Validation (XSS)	WebView script execution	Static (jadx)	Captured

Optional: to also show the in-app scoreboard, enter these values in BeetleBug on an emulator/device and add the scoreboard screenshot below.



6. Conclusion

Every BeetleBug challenge category was solved and its flag or secret recovered. The recurring root causes are consistent with the OWASP Mobile Top 10: secrets and sensitive data trusted to the client and device (C1, C2, C3), components and interfaces exported without authorisation (C4, C5, C6, C7), a misconfigured cloud backend (C8), and unvalidated input reaching a query or a WebView (C9, C10). The single Critical issue — the world-readable Firebase database — exposed live data without any authentication.

Applying the per-challenge remediations — server-side secret management, Keystore-backed encrypted storage, stripped release logging, restricted exported components, hardened WebViews, Keystore-bound biometrics, authorised deep links, locked-down Firebase rules, and parameterised queries — would close all demonstrated issues and align the app with the OWASP MASVS.

End of report — BeetleBug Mobile Application Security Assessment, version 1.0, 6 October 2026. Educational use.